



A trans-species cytoplasmic polymorphism is associated with seed shape and aridity across multiple species of sunflowers

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The cytoplasmic genomes of plants and animals often fail to track species boundaries. However, the mechanisms responsible for such patterns are poorly understood, in part because few studies have linked cytoplasmic variation to phenotypic traits or environmental differences. Here, we use 1,554 previously published and 185 new whole genome sequences representing 14 taxa from the sunflower genus *Helianthus*, 91 phenotypic traits measured in common gardens, and 39 environmental variables to test for environmental and phenotypic effects of cytoplasmic genome variation. In agreement with previous work, two distinct chloroplast clades were found across multiple species and the sharing of chloroplast clades between species was mainly due to repeated introgression rather than incomplete lineage sorting. Two mitochondrial clades were also found that matched the chloroplast clades for 98% of individuals, implying predominantly maternal inheritance of both genomes. Cytoplasmic clade was associated with differences in seed shape across several species, and likely with aridity, suggestive of a role in local adaptation. Conversely, we failed to find any credible cytonuclear interactions based on associations between chloroplast and nuclear variation. Taken together, this work suggests that cytoplasmic genomes in annual sunflowers represent a trans-species balanced polymorphism that is likely maintained by adaptation to different environments. More broadly, our results corroborate the syngameon concept, showing how introgression across even very strong reproductive barriers can facilitate environmental adaptation across a species complex.

cytoplasmic genomes | introgression | local adaptation | sunflower | trans-species polymorphism

Early molecular phylogenetic studies made a surprising discovery: namely that phylogenies of cytoplasmic genes or genomes were often discordant with those based on nuclear genes. This discovery hinted at the possibility that gene flow across species barriers (introgression) was more common than previously believed (1, 2). However, introgression was difficult to prove because the sharing of alleles between species can also result from incomplete lineage sorting (ILS), which occurs when ancestral variation is maintained across a speciation event (3, 4). Thus, at an ILS locus, an individual may be more closely related to a different species than to individuals of its own species at that part of the genome, similar to the pattern produced by introgression. Beyond concerns about ILS, it was unclear whether the observation of frequent sharing of cytoplasmic genomes extended to the nuclear genome. These questions have been resolved over the last two decades due to the accumulation of next-generation sequencing data for many species and the development of new analytical methods for distinguishing between introgression and ILS (5–8). The results are clear: Introgression is common during the evolution of many plant and animal groups (9). These findings have revitalized the syngameon concept, which describes groups of otherwise distinctive species that are linked together by hybridization and introgression (10–12). Syngameons have been suggested to lower demographic and genetic risks of extinction, while also offering a means for accelerating adaptation by permitting the introgression of beneficial mutations (13, 14).

While cytonuclear or gene tree discordance can be neutral, selection can also modify its frequency. For example, balancing selection extends the amount of time variation is maintained, leading to more frequent discordance (15). Introgressed alleles with effects on fitness may be removed due to negative selection against the allele (or linked sites) when in a different environment or genetic background, but selection favoring introduced alleles can lead to their retention and eventual increase in frequency within the recipient species (i.e. adaptive introgression) (12).

Significance

Plant cells have three genomes. The largest of these is housed in the nucleus, whereas smaller genomes are found in chloroplasts and mitochondria, which reside in the cytoplasm. Evolutionary histories based on nuclear genes sometimes disagree with those from cytoplasmic genes. In this paper, we uncover the evolutionary mechanisms responsible for this pattern in annual sunflowers. We find that multiple sunflower species share two divergent cytoplasmic lineages due to interspecific hybridization and that the divergent cytoplasmic underlie differences in seed shape and likely adaptation to several environmental factors. Thus, despite having distinctive morphologies and karyotypes, sunflower species evolve as a network of adaptive alleles, emphasizing that connectivity via gene flow contributes to the persistence of species on fragmented landscapes.

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To better understand how selection is affecting patterns of introgression or ILS, we need knowledge about the genetic underpinning of adaptation (16–18). Such knowledge can be gained through genome-first approaches, which look for the signatures of natural selection on genetic variation (19, 20). Alternative approaches have focused on linking genetic variation to phenotypes, either known to, or plausibly inferred to, be under selection using genome-wide associations or QTL mapping (21, 22). Last, genome-environment approaches search for alleles associated with environmental variables that may influence fitness and promote local adaptation (23, 24). These genomic approaches typically focus on the nuclear genome and exclude cytoplasmic genomes from the analyses. While dramatically smaller than nuclear genomes, the mitochondrial and chloroplast genomes contain genes critical to respiration and photosynthesis, respectively, and cytonuclear interactions appear to be common (25, 26). Also, due to uniparental inheritance and limited recombination between cytoplasmic genomes of different individuals (27), cytoplasmic genomes typically act as a single large effect locus. These properties create the potential for cytoplasmic genome variation to have disproportionately large effects on adaptation and fitness compared to individual nuclear loci, suggesting cytoplasmic genomes should not be ignored when exploring adaptation (28).

Evidence for adaptive cytoplasmic variation has come from several approaches. Phylogenetic studies testing the rate of coding evolution in organellar genes have often found variation consistent with positive selection, although links with functional significance are rare (29–32). Quantitative genetic approaches have shown that cytoplasmic variation has large effects on the metabolome (28), as well as growth and fitness-related traits (33, 34). The relative paucity of recombination in cytoplasmic genomes can also lead to the accumulation of deleterious alleles and subsequently the turnover of lineages (26, 32). A common theme in many of these studies concern epistatic interactions between organellar and nuclear genome variation, also known as cytonuclear interactions. These are especially common in the context of hybrids between species and can be powerful reproductive barriers (35). Despite this, cytonuclear discordance, where closely related organellar haplotypes are shared across species that otherwise show divergence across their nuclear genomes, is common and represents a kind of trans-species polymorphism (36–40).

The annual sunflowers of the genus *Helianthus* are a useful system for testing the drivers of cytoplasmic trans-species polymorphism. Studies of the nuclear genome have found evidence of introgression between many species in the annual sunflower clade despite strong reproductive barriers (41–46). Early studies detected some level of cytonuclear discordance (47–50) and more recent work revealed two main chloroplast clades that are shared among species within the group (51). Lee-Yaw et al. found that the phylogenetic pattern for the chloroplast genome was most likely due to introgression, rather than ILS, and that there was evidence for positive selection based on substitution patterns. Reciprocal transplant experiments of segregating backcross populations suggested that the two chloroplast clades may be adapted to different environmental (i.e. soil) conditions (52). However, the authors were unable to disentangle the phenotypic and environmental factors responsible for differential fitness in the transplant gardens (52). Furthermore, little is known about mitochondrial genome variation in this group, and associations between cytoplasmic and phenotypic variation remain to be explored.

Here, we analyzed a large dataset of sequenced *Helianthus* and associated phenotypic data from an exhaustive common garden study (> 90 phenotypic traits recorded) to determine the extent of chloroplast and mitochondrial DNA discordance across all

sequenced species and to explore the evolutionary mechanisms responsible for the patterns observed. In particular, we conduct tests to distinguish between introgression and ILS, search for phenotypic and environmental effects of cytoplasmic clade, and test for cytonuclear interactions.

Results

A Trans-species Cytoplasmic Polymorphism in Annual Sunflowers. To understand the distribution of cytoplasmic genome variation in annual sunflowers, we analyzed the whole-genome sequence data first reported by Todesco et al. (53), Owens et al. (54), Bercovich et al. (55), and Huang et al. (56) (Dataset S1). This collection was largely focused on wild *H. annuus* (n = 729), *H. argophyllus* (n = 299), *H. petiolaris petiolaris* (n = 168), *H. petiolaris fallax* (n = 241), and *H. niveus canescens* (n = 86). Additionally, we included a smaller number of samples from the annual sunflowers *H. debilis* (n = 8), *H. paradoxus* (n = 4), *H. anomalus* (n = 178), and *H. deserticola* (n = 9), of which 185 are presented here for the first time. Last, we included samples from the perennial sunflowers *H. divaricatus* (n = 9), *H. giganteus* (n = 4), *H. decapetalus* (n = 1), *H. grosseserratus* (n = 3), and *Phoebanthus tenuifolius* (n = 1). The annual sunflowers are monophyletic, so the perennial sunflower species can act as an outgroup, while *Phoebanthus* is the sister genus to *Helianthus* and acts as a more distant outgroup. While most of the samples included have been previously analyzed for their nuclear genome, cytoplasmic genome variation was largely ignored.

Despite treatment with a Duplex-Specific Nuclease to reduce the proportion of repetitive sequences in our libraries (53), we achieved relatively high depth for both cytoplasmic genomes (mean chloroplast depth = 225, mean mitochondrial depth = 90). Phylogenetic trees reconstructed using IQ-TREE divided most samples into two clades for the chloroplast and mitochondrial datasets (Fig. 1B and SI Appendix, Fig. S1). For the chloroplast tree, we found two groups of samples that fell at the base of the clade 2 with short internode branch length. To understand whether these samples were highly diverged clade 2 samples or an ancestral clade, we identified sites that had >95% difference in allele frequency between clades 1 and 2 and designated these as diagnostic loci. Based on comparisons to *P. tenuifolius*, we designated each allele as derived or ancestral, thereby producing a list of derived alleles characteristic of each clade. We found that the basal samples had the ancestral allele for 24/24 of clade 1 diagnostic derived sites and 15/16 of clade 2 diagnostic derived sites. We classified all of these samples as other lineages since they were nonmonophyletic and generally, but not exclusively, basal to the two named clades. Together, 1.3% of samples in the annual species group were classified as other lineages.

All but one of the perennial samples fell outside of clades 1 and 2. Some grouped close to the *Phoebanthus* root while others were just basal to clades 1 and 2. One *H. divaricatus* sample had the clade 2 cytotype. Considering all other perennial sunflowers grouped with the outgroup, we expect this is due to introgression, rather than suggesting clade 2 was ancestral.

We examined pairwise genetic distances derived from our IQ-TREE analysis, excluding all outgroup and perennial sunflower samples, between our chloroplast clade groupings (Table 1 and SI Appendix, Fig. S2). We found the lowest distance was between samples in clade 2 and that the highest distance was between samples classified as other lineages. Furthermore, there was greater average distance between two samples in the other lineages group, than between clade 1 and clade 2, suggesting that the other lineages bin includes multiple diverse lineages.

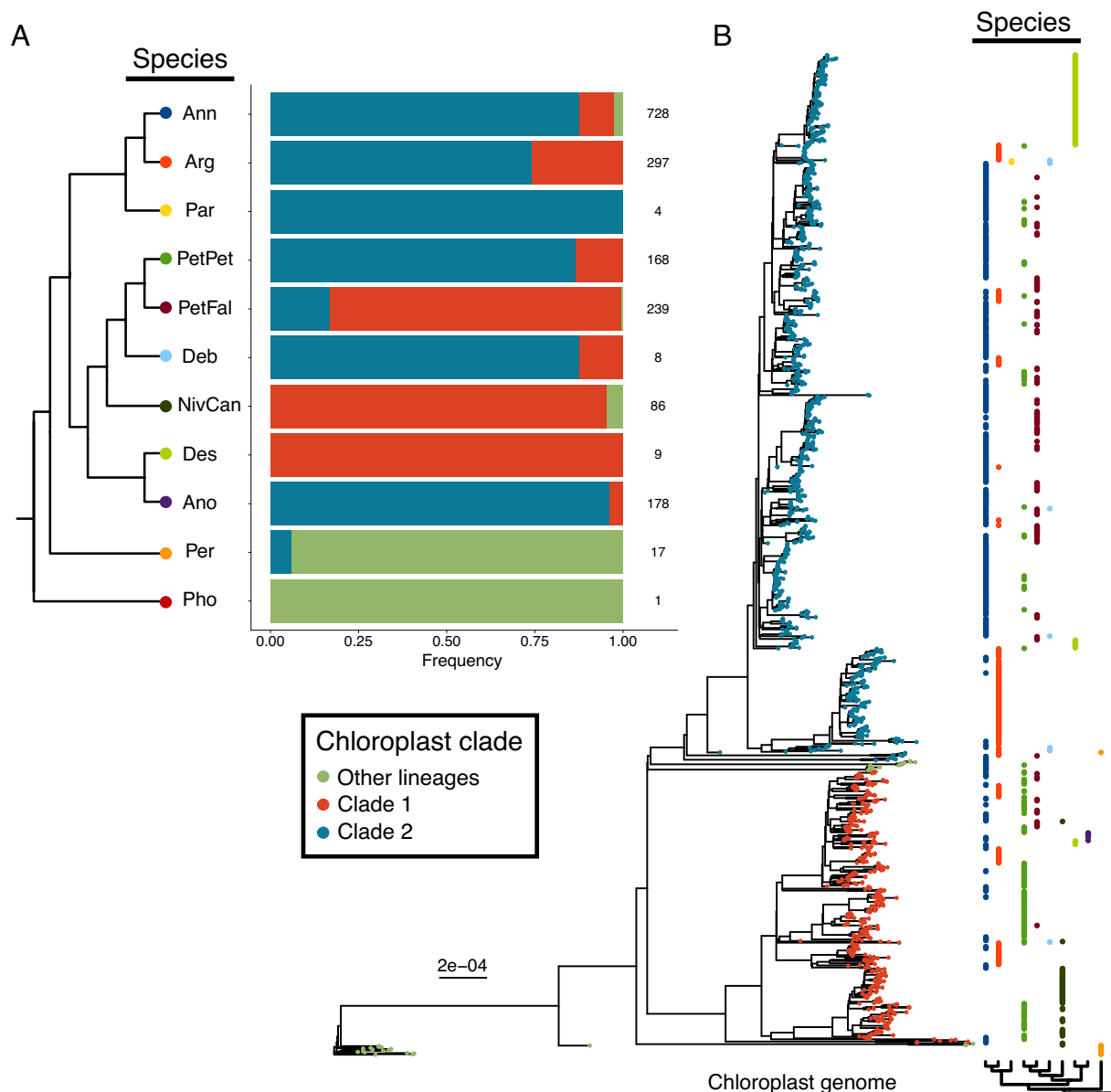


Fig. 1. Chloroplast genome phylogeny and the proportion of samples with each chloroplast clade. (A) Proportion of samples from each species with each chloroplast clade. (B) Maximum likelihood phylogeny of chloroplast genome reveals two clades across multiple species. The chloroplast clade is coded in red (clade 1), blue (clade 2), or green (other lineages). The species identity is color coded after the tip and is ordered by the species tree topology. All perennial *Helianthus* species are grouped together into a single color code. Ann: *H. annuus*, Arg: *H. argophyllus*, Par: *H. paradoxus*, PetPet: *H. petiolaris petiolaris*, PetFal: *H. petiolaris fallax*, Deb: *H. debilis*, NivCan: *H. niveus canescens*, Des: *H. deserticola*, Ano: *H. anomalus*, Per: All perennial *Helianthus*, Pho: *Phoebanthus tenuifolius*. The species topology is taken from Owens et al. 2023.

We found that although each species had a single dominant chloroplast clade, alternate clades were common and found in all species with >10 individuals sampled (Fig. 1A). Additionally, the topology of the clades did not match the species topology. For example, the *H. petiolaris* subspecies differed at their dominant clade, as did the sister species *H. anomalus* and *H. deserticola*. Within a single chloroplast clade, species were not monophyletic (Fig. 1B). For example, for *H. petiolaris petiolaris*, *H. petiolaris fallax*, *H. argophyllus*, and *H. debilis*, most individuals fell within parts of the tree that were mainly *H. annuus*. Although *H. anomalus* is primarily a single monophyletic clade, this species has a limited range and samples were mostly from only two populations, in contrast to the wide sampling range of the other species included.

For the mitochondrial genome, we found highly similar patterns; two clades along with a minority of seemingly basal samples, classified as other lineages (SI Appendix, Fig. S1). We again found

that the other lineages did not contain derived alleles diagnostic for the two clades, supporting that they were not recombinant lineages. The clade identity between the chloroplast and mitochondrial genomes overwhelmingly agreed but were found to be cytoplasmic mismatched in 1.9% of samples (SI Appendix, Fig. S3). Despite cytoplasmic mismatching occurring in 32 individual samples, they clustered in 6 positions in the tree, suggesting biparental inheritance is rare. Given the agreement between the two cytoplasmic genomes, we would have little power to distinguish between effects of each individual cytoplasmic genome. Therefore, we focused our remaining analyses using clade identity from the chloroplast genome, which had greater resolution, but recognize that any effects may be driven by either cytoplasmic genome.

For all species, there was a single majority cytotype. We designated samples with the minority cytotype for their species as nuclear mismatched, to distinguish them from cytoplasmic

Table 1. Median pairwise genetic distance calculated by IQ-TREE between the different chloroplast clades, using model K3Pu+F + I + R4

Group 1	Group 2	Median distance	SD
Clade 1	Clade 1	0.0000934	0.000147
Clade 2	Clade 2	0.0000258	0.0000661
Clade 1	Clade 2	0.000206	0.0000929
Other lineages	Clade 1	0.000403	0.000163
Other lineages	Clade 2	0.000450	0.000146
Other lineages	Other lineages	0.000502	0.000307

mismatch which occurred between chloroplast and mitochondrial genomes. Nuclear mismatched samples were not limited geographically but were found in multiple populations for each species (Fig. 2). This mismatch may be due to recent introgression between species, which often occurs (46, 57). To test this, we asked whether the closest relative of the mismatched lineage (e.g. a possible donor lineage) was geographically closer than other possible donor lineages. Likely donors were geographically closer than expected for *H. annuus* ($t_{29} = -7.23$, $P = 5.82E-08$), *H. petiolaris fallax* ($t_{36} = -4.43$, $P = 8.34E-05$), and *H. petiolaris petiolaris* ($t_{21} = -5.41$, $P = 2.29E-05$) but not for *H. argophyllus* ($t_{42} = 1.41$, $P = 0.15$) (SI Appendix, Fig. S4A). The latter is native to the coastal plain of Texas, and its restricted geographic distribution may have reduced the power of this analysis. To test whether our analysis is robust to node support, we collapsed all nodes with < 50% ultra-fast bootstrap support and repeated this analysis and found qualitatively the same results (SI Appendix, Fig. S4B).

We then asked whether populations with nuclear mismatch had more nuclear introgression than others using Patterson’s D with all pairs of populations. Since Patterson’s D compares two focal populations with a third donor/receiver population, it estimates only relative shared ancestry between the two focal populations, not an unbiased estimate of introgression. For example, if two populations are both introgressed with the same amount of donor ancestry, then Patterson’s D would likely find no evidence

for introgression (58). By taking the average D for each population with all possible pairs, we rank each species by relative introgression amount. We found the strongest evidence for this hypothesis in *H. petiolaris*; nuclear mismatched *H. petiolaris fallax* populations are more admixed with all possible donors, but most strongly from *H. petiolaris petiolaris* (SI Appendix, Fig. S5). There is a non-significant enrichment of introgression signal in *H. petiolaris fallax* from *H. petiolaris petiolaris* for nuclear mismatched populations. This suggests that nuclear mismatch may be driven by recent admixture between the *H. petiolaris* subspecies, but not for other species.

Possible Adaptive Explanation for Cytoplasmic Polymorphism.

The spread of two clades across multiple species boundaries suggests the action of gene flow, potentially coupled with natural selection. Previous experimental analyses have found signs of local adaptation in cytoplasmic genomes when crossing *H. annuus* and *H. petiolaris fallax* in New Mexico (52). We explored this hypothesis by testing for significant associations between the proportion of clade 1 chloroplast genomes per site and 24 topoclimatic and 15 soil variables in five of the species: *H. annuus*, *H. argophyllus*, *H. niveus canescens*, *H. petiolaris petiolaris*, and *H. petiolaris fallax*. We first ran simple, univariate tests, testing the effect of each environmental variable on chloroplast clade proportion, with or without controlling for species. To explore multivariate patterns, we used a principal component analysis (PCA) of all variables to identify the top axes of environmental variation, and ran linear models using the top five principal component axes as predictors, again with and without controlling for species. Last, although our population sample size is insufficient to confidently identify patterns within species, we repeated the association test by dividing our data by species and tested each environmental variable independently per species to see whether all species trended in the same direction.

We found several climate and soil variables were significantly associated with the cytotype proportion, but this association was largely removed if species was included in the model (Table 2). Before inclusion of species, the most significant association was with Hargreaves climatic moisture deficit, soil organic matter, and

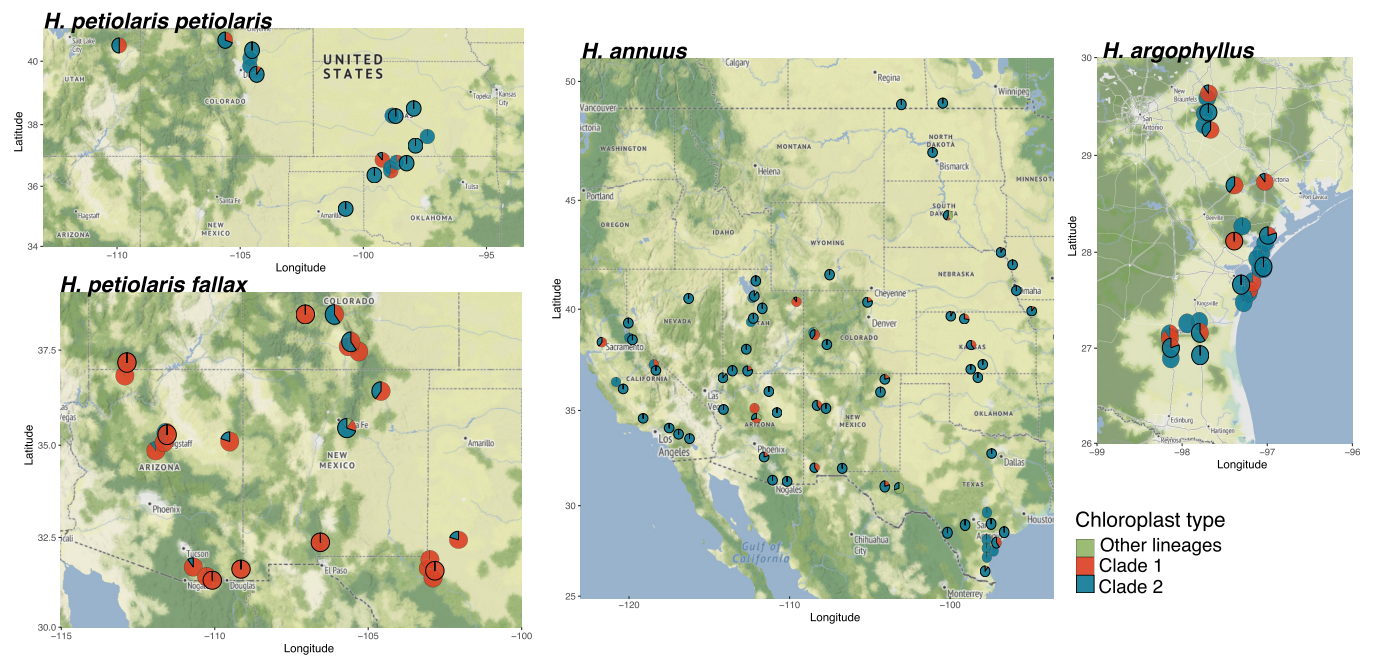


Fig. 2. Mismatched cytoplasmic types are distributed across the species range. Map background colors show terrain from the Stamen Terrain style of Stadia Maps.

Table 2. Chloroplast environment associations as a general linear model, with and without controlling for species

Variable	PVE	PVE ^{species}	P	P ^{species}	P ^{adj}	P ^{adj} ^{species}	Variable type
CMD	9.44	0.73	1.67E-05	1.53E-01	6.50E-04	1	Combined
OM	8.17	0.42	2.23E-04	7.39E-01	8.71E-03	1	Soil
CEC	8.16	0.38	1.85E-04	9.69E-01	7.20E-03	1	Soil
RH	7.87	1.86	1.27E-04	2.15E-02	4.97E-03	0.84	Precipitation
Elevation	7.21	0.21	2.10E-04	4.39E-01	8.19E-03	1	Location
CA	7.08	0.42	6.28E-04	7.25E-01	2.45E-02	1	Soil
SOL_SALTS	6.67	0.38	1.23E-03	9.42E-01	4.80E-02	1	Soil
MAP	6.40	0.36	5.11E-04	3.16E-01	1.99E-02	1	Precipitation
NA	6.11	0.38	1.94E-03	9.83E-01	7.55E-02	1	Soil
MG	4.46	1.46	1.25E-02	8.09E-02	4.87E-01	1	Soil
K	4.35	0.38	1.26E-02	9.00E-01	4.92E-01	1	Soil
Eref	3.64	0.48	9.16E-03	2.53E-01	3.57E-01	1	Combined
MSP	3.27	0.40	1.30E-02	2.85E-01	5.08E-01	1	Precipitation
BICARB	2.71	0.63	8.17E-02	4.04E-01	1	1	Soil
PERCENT_NA	2.47	0.88	1.09E-01	2.42E-01	1	1	Soil

PVE: Percent variance explained by the variable. Species: models including species as a covariate. Adjusted P-values are corrected using Bonferroni correction. CMD: Hargreaves climatic moisture deficit, OM: Soil organic matter, CEC: Soil cation exchange capacity, RH: Relative humidity, CA: Soil calcium, SOL_SALTS: Soil soluble salts, MAP: Mean annual precipitation, NA: Soil sodium, MG: Soil magnesium, K: Soil potassium, Eref: Hargreaves reference evaporation, MSP: Mean summer precipitation, BICARB: Soil bicarbonates, PERCENT_NA: Soil percent sodium. Bolded values indicate $P < 0.05$.

cation exchange capacity; after inclusion of species, the strongest association was with relative humidity, although this association was not significant after correcting for multiple testing. Similarly, when testing for associations with principal components and cyto-type proportion we found that PC3, primarily driven by soil and moisture variables, was associated with cytotype proportion, but only if species was not controlled (Fig. 3 and Dataset S3). When each individual species was tested independently, we failed to find consistent signals of effects across all species, although relative humidity showed a positive relationship in three out of four species (SI Appendix, Fig. S6). We repeated these analyses using mito-chondrial clade instead of cytoplasmic clade and found similar results (Datasets S2 and S3).

We explored whether cytoplasmic clade affected variation in 91 phenotypic traits measured in common gardens for *H. annuus*, *H. argophyllus*, *H. petiolaris petiolaris*, and *H. petiolaris fallax* (53, 55). Many of the traits were correlated (for example, different leaf shape measurements) so we used a PCA to summarize the major axes of variation in architecture, inflorescence, leaf, and seed traits. We then asked whether cytoplasmic clade was associated with variation in the first two principal components of each trait type (SI Appendix, Figs. S7–S10). We found significant associations with seed PC1, which explains 47.6% of variation ($P = 0.04$), and PC2, which explains 26.0% of the variation ($P = 0.00008$) (Fig. 4A and Dataset S4). The association with PC2 remained significant after Bonferroni correction for multiple testing ($P = 0.0006$). To understand what aspect of the seed morphology was being affected by the cytoplasmic genome, we tested each seed trait individually and found significant associations with 13 traits, largely related to whether a seed was short and round, or long and narrow (Fig. 4B). When using mitochondrial clade, we found slightly stronger associations with seed shape PCs (Dataset S4).

Another way for cytotype to play an adaptive role is through epistatic interactions with the nuclear genome. We tested this by looking for correlations between nuclear allele frequencies and cytotype frequency in populations, and then finding windows enriched for low P -values using WZA (59). While SNP-based methods look for exceptional correlations between SNP genotype and other variables, WZA looks for genomic windows that have

higher proportions of low P -values. This allows for the detection of candidates where no individual SNP is globally significant, but the signal across multiple SNPs suggests selection. Our initial results showed no evidence of individual peaks of association for SNPs in *H. annuus* and both *H. petiolaris* subspecies, but some-what higher values in *H. argophyllus*, possibly due to the larger proportion of mismatched samples (SI Appendix, Fig. S11). We then looked for windows with disproportionate numbers of low P -values using WZA, and found much more evidence of signifi-cant peaks (SI Appendix, Fig. S12). In particular, we find high WZA scores on chromosome 13 in *H. annuus*, corresponding to a previously identified haploblock (53). Despite this, the tail of the P -value distribution was not enriched for low scores compared to permuted tests for either SNP or window statistics, suggesting the peaks we observed could occur by chance, and we do not have convincing evidence for cytonuclear associations (SI Appendix, Figs. S13–S14).

Discussion

Cytotype Sorting Across the Annual Sunflowers. As previously reported by Lee-Yaw et al. (51), the distribution of cytoplasmic clade does not match the overall species tree topology. At the species tree level, cytoplasmic clade 1 is predominant in *H. petiolaris fallax*, *H. niveus canescens*, and *H. deserticola*, but not the closely related *H. debilis* or *H. petiolaris petiolaris*. Conversely, Clade 2 is predominant in *H. annuus*, *H. argophyllus*, and *H. paradoxus*, but not *H. bolanderi*, which is sister to *H. annuus* (results for *H. bolanderi* from (51)). Based on sequence divergence, Lee-Yaw et al. estimated that the clades diverged up to 250 KYA, and are younger than all of the species involved (41, 60). If this is accurate then the two clades have only recently swept across the annual sunflowers replacing ancestral variation. This is consistent with our finding that the median genetic distance was highest between samples in the other lineages, suggesting it includes multiple lineages of remnant chloroplast variation. This also explains the disagreement between species topology and cytoplasmic clade. It is unclear which species the two clades originated in, but since there are many more species than clades, the dominant cytoplasmic clade

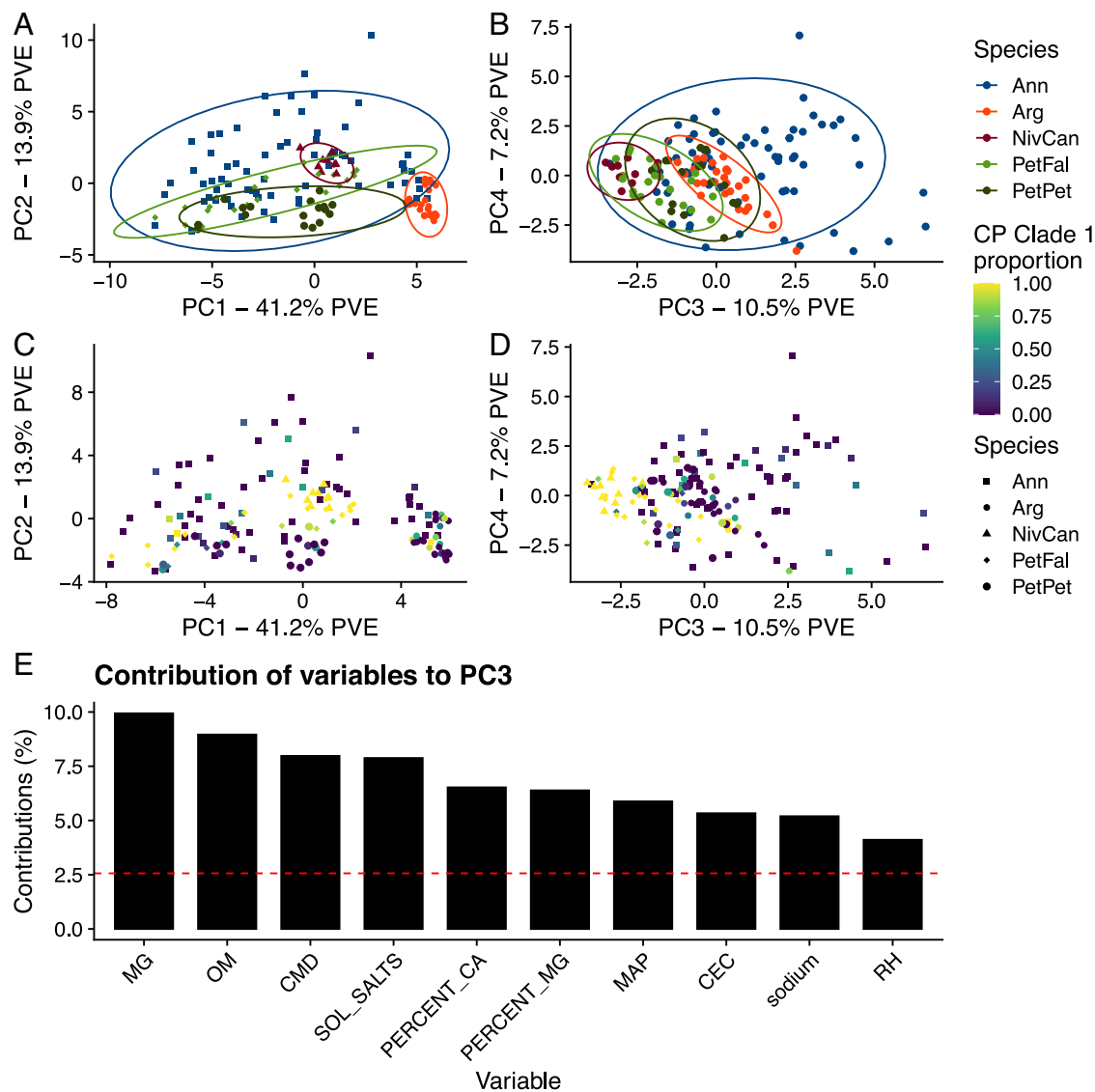


Fig. 3. PCA of soil and climate variables for annual sunflower populations. (A and B) Principal components 1 to 2 and 3 to 4 with populations and 95% ellipses color coded by species. (C and D) Principal components 1 to 2 and 3 to 4 with populations color coded by chloroplast clade proportion and species coded by shape. (E) The top 10 variables contributing to PC 3, which is associated with chloroplast clade. The red dotted line represents the average contribution across all variables. Ann: *H. annuus*, Arg: *H. argophyllus*, NivCan: *H. niveus canescens*, PetFal: *H. petiolaris fallax*, PetPet: *H. petiolaris petiolaris*. MG: Soil magnesium, OM: Soil organic matter, CMD: Hargreaves climatic moisture deficit, SOL_SALTS: Soil soluble salts, PERCENT_CA: Soil percent calcium, PERCENT_MG: Soil percent magnesium, MAP: Mean annual precipitation, CEC: Soil cation exchange capacity, NA: Soil sodium, RH: Relative humidity.

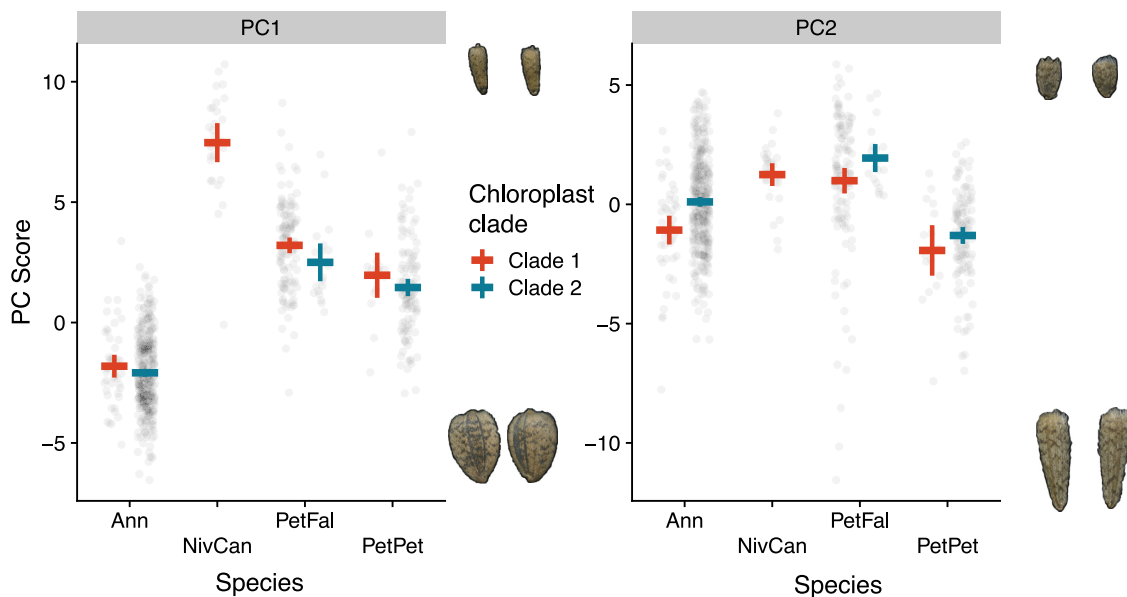
seems to have arrived by recent replacement of ancestral variation in most species. It is possible that the choice of dominant cytotypic was dependent on environmental and ecological selection, rather than phylogenetic history or evolutionary contingency. The relatively young age of the cytoplasmic clades suggests that it may be in the goldilocks zone, enough evolutionary time to adaptively differentiate but not enough for strong cytoplasmic incompatibilities to evolve. Interestingly, we find two cases where sister species (or subspecies) have contrasting cytotypes. It is unlikely that the cytotypic directly played a role in reproductive isolation, given that nuclear mismatches are common. We think it is more likely that both the speciation process and the choice of predominant cytotypic were driven by ecological adaptation.

We tested whether nuclear cytoplasm mismatch was due to recent introgression using two approaches. Together they suggest that mismatched cytoplasmic genomes for most species are introgressed from geographically close donors, but populations with more mismatches are not exceptionally admixed, excluding *H. petiolaris fallax*. We

think this means that mismatched cytoplasmic genomes came from introgression since populations spread to their current geographic regions, but far enough in the past for most nuclear admixed ancestry to be purged. The repeated introgression of cytoplasmic genomes across species is further supported by the phylogenetic tree, which shows that even within a clade, species are rarely monophyletic, and by previous population simulation-based tests (51).

Surprisingly, we find that the mitochondrial and plastid clade identity disagrees in 1.9% of samples. This indicates past cases of biparental inheritance of cytoplasmic genomes, in contrast to typical maternal inheritance. Although a study in *H. annuus* found no paternal leakage, further work in *H. verticillatus* found paternal leakage in 1.9% of offspring (61, 62). Paternal leakage initially produces cytoplasmic heteroplasmy, which could lead to replacement of one cytoplasmic genome but not the other and the pattern we observe (63). Paternal leakage rates likely vary between crosses and species, so it is unclear what the expected amount of mismatch in our dataset is in the absence of selection. In this way, the

A Seed traits



B Effect size of chloroplast clade on seed traits

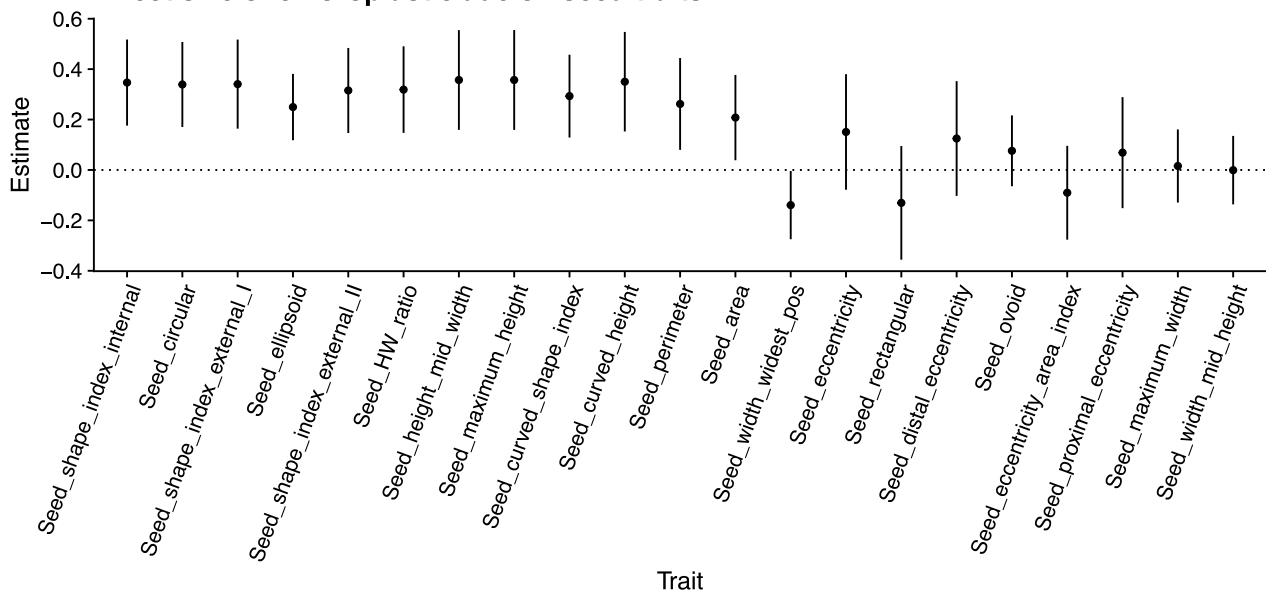


Fig. 4. Cytoplasmic type affects seed traits across species. (A) The distribution of principal components 1 and 2 for each species and chloroplast clade. Solid lines represent mean and 95% CI for the mean. Example seed images are shown for the most extreme values of each PC axis. (B) Mean and 95% CI for the effect size of chloroplast clade on seed traits, controlling for species. Traits are ordered by *P*-value.

mitochondria and chloroplast genomes are like two closely linked markers, with a small, but nonzero recombination rate. Depending on what this rate is, the amount of mismatch may reflect neutral sorting, or coadaptation and selection keeping the two haplotypes together. Future work could utilize populations with high levels of cytoplasmic mismatch in a controlled cross experiment to test for the effects of individual cytoplasmic genomes.

Phenotypic and Environmental Cytotype Associations. Why are two cytotypes maintained in the annual sunflowers? Previous work used a backcross design to exchange cytoplasmic genomes between *H. annuus* and *H. petiolaris* (52). They found that the *H. annuus* cytoplasm, likely clade 2, was superior in mesic environments, while the *H. petiolaris* cytoplasm, likely clade 1, was superior in xeric environments. In our dataset, we see associations with moisture and soil variables and chloroplast cytotpe, but this effect is confounded

by environmental differences between species. Given that the two cytotypes are likely recently acquired, this suggests that the majority cytotpe may be driven by environmental matching. In other words, cytotpe–environment interactions may occur at the species level, more than the population level. We tested this here using environmental data for five species; future work could test for associations across all annual sunflower species. After controlling for species, the strongest association we observe is with relative humidity, consistent with the experimental work linking cytotpe to water availability and the effect is in the same direction. Although the effect is not significant after controlling for multiple testing, we think this is suggestive considering the previous experimental work, and is worth following up with experimental tests of water use efficiency or drought tolerance and cytotpe. Ecological adaptation by cytoplasmic genomes has been seen in *Nothofagus*, where chloroplast sequences group by latitude rather than species identity

across five species in South America (36), and in *Brassica*, where weedy *B. rapa* have preferentially captured the *B. napus* chloroplast (64). Future experimental work could test for differences in salinity or temperature tolerance between cytotypes.

Cytotype was also found to have a consistent effect on seed shape across multiple sunflower species, with the clade 1 cytotype tending to be associated with larger, longer seeds than the clade 2 cytotype. Cytoplasmic variation has been shown to affect seed traits in other species, including maize and oilseed rape (65, 66). Seed shape is an ecologically important trait. In particular, in *H. petiolaris fallax* and *H. neglectus* (which genetically falls within *H. petiolaris fallax*), dune adapted populations have significantly larger and longer seeds (67). It is hypothesized that larger seeds give seedlings enough resources to emerge from buried sand or prevent burying. From our dataset, we include two paired dune/nondune populations of *H. petiolaris fallax* that have large differences in seed size and shape, as well as large regions of differentiation across the nuclear genome (67, 68). In the Monahan's Sandhills State Park region, both dune and nondune populations are uniformly clade 1 cytotype. For the Great Sand Dunes National Park region dune and nondune populations were polymorphic for cytotype and dune populations were predominantly clade 2 while nondune population were predominantly clade 1, contrary to our expectations based on seed shape. It is possible that selection on seed size and selection on environment are in opposing directions, leading to unpredictable responses.

A Lack of Signal for Cytonuclear Interactions. Coevolution between nuclear and cytoplasmic genomes can result in negative epistatic interactions, which can drive the replacement of cytoplasmic genomes during hybridization (35, 69, 70). We searched for cytonuclear interactions using an approach similar to a genome-environment association and used a correlation coefficient, Kendall's tau, because mismatch samples were not generally geographically structured, unlike genome-wide population structure. Additionally, work by Booker et al. suggested little benefit would be gained from the application of more complicated methods (59).

Although the initial WZA showed several regions with significant *P*-values, there was little difference in the tail of the *P*-value distribution between empirical and permuted cytotype scores. This means we are unable to confidently find true positives for cytonuclear interactions. We suspect that regions of high LD in the sunflower genome lead to significant WZA regions, even in the absence of a true association.

A lack of credible cytonuclear interactions is perhaps not surprisingly given the recent age of the cytoplasmic clades (51). Most likely, neither clade has evolved long enough in a single nuclear background to evolve specific genetic interactions. Additionally, our findings of the link between cytotype and environment suggest that cytonuclear interactions may be mediated through the environment. Rather than direct interactions between cytoplasmic and nuclear proteins, genetic variation in both may contribute to adaptation to the same environmental variables. This would result in linkage between the genomes in the wild, but experimental fitness effects would be environmental dependent. If this is the case, then we expect that any cytonuclear interactions to be highly diffuse across the nuclear genome reflecting the polygenic nature of environmental adaptation.

Conclusions

Taken together, our work suggests that cytoplasmic genomes act like a pleiotropic trans-species polymorphism that is under balancing selection across environments. This paints a picture of the

annual sunflowers acting as a syngameon, allowing adaptive alleles to flow across species boundaries (13). The sharing of environmentally associated genomic regions is also found in the nuclear genome, although whether such sharing is due to introgression versus ILS remains unclear (71). Uniparental inheritance and lack of recombination between the cytoplasmic genomes from different individuals leads to strong linkage disequilibrium within and between cytoplasmic genomes, resulting in combined phenotypic effects similar to that seen for inversions (72). This complicates the selective landscape and likely constrains cytoplasmic derived local adaptation (73).

Although the application of the syngameon concept has mainly been restricted to plants, such hybridizing networks are also common in animals (13). For example, complex hybridization appears to have facilitated adaptive radiations in cichlid fishes (74) and the origin of new hybrid lineages in *Heliconius* butterflies (75). Additional potential benefits from participating in syngameons include increased diversity (76), as well as access to novel allelic and phenotypic diversity (12). It has been speculated that such benefits could reduce extinction rates during periods of rapid environmental change (77), including Anthropogenic climate warming (78, 79). However, hybridization can lead to the local extinction of populations through genetic or demographic swamping (80). The harms of maladaptive gene flow can be reduced by prezygotic reproductive barriers, which are strong in sunflower (41, 44) and a number of other well-characterized syngameons, such as the Louisiana irises (81) and European white oaks (82). Postzygotic reproductive barriers also reduce the likelihood of genetic swamping, but they can increase vulnerability to demographic swamping if hybridization rates are high (83). However, demographic swamping appears to be rare (80). Thus, our work supports recent arguments that the benefits of introgression in most syngameons likely outweigh potential harms (79, 84).

Materials and Methods

Processing of Sequence Data and Phylogenetic Analyses. We analyzed 1739 whole genome sequences representing 14 taxa from the sunflower genus *Helianthus* (Dataset S1). The samples were largely previously published (46, 53, 55), but included additional samples for three species: *H. paradoxus* (*n* = +2), *H. anomalus* (*n* = +176), and *H. deserticola* (*n* = +7). Common garden trait data and environmental data were available for most samples from *H. annuus*, *H. argophyllus*, *H. petiolaris petiolaris*, *H. petiolaris fallax*, and *H. niveus canescens*.

For each sample, raw data were trimmed using Trimmomatic (v0.22) (85), and aligned to only the chloroplast and mitochondrial genomes of *Helianthus annuus* (NCBI KF815390.1 & DQ383815.1) using NextGenMap (v0.5.5) (86). Variants were then called using samtools (v1.10) *mpileup* and bcftools (v1.10.2) *call* assuming a haploid state for individual samples for the entire cytoplasmic genome, and then merged together into a single VCF (87). Individual genotypes were filtered to require at least 10 reads for each position. Last, the VCF, with indels removed, was converted into an aligned fasta file for the chloroplast and mitochondrial genomes by a custom perl script which excludes indels. While indels may be functionally important, we focused our analysis here on clade identity, rather than individual mutations, and indels are more challenging to genotype with short read data.

We used IQ-TREE with the model finder and 1,000 ultrafast bootstraps to generate separate chloroplast and mitochondrial maximum likelihood phylogenies (88–90). Our chloroplast tree revealed the two major clades observed by Lee-Yaw et al. (51) along with a third polyphyletic group we named other lineages (*Results*) and we used membership in these clades to assign a chloroplast haplotype (clade 1, clade 2, other lineages) to individuals. We likewise assigned individuals to one of three, corresponding, mitochondrial haplotypes. We found overwhelming agreement between chloroplast and mitochondrial clade assignment for individuals, so for most analyses, we focused on the chloroplast genome.

Testing for Recent Cytoplasm Introgression. Mismatches between cytoplasmic and nuclear genome can be caused by ILS or introgression. In our study, each species had one dominant chloroplast/mitochondrial clade, and in some cases contained a minority of individuals with the other clades. For this analysis, we focused only on mismatches between clade 1 and clade 2 because the other lineages were rare in all annual species. Additionally, we only focused on *H. annuus*, *H. argophyllus*, *H. petiolaris petiolaris*, and *H. petiolaris fallax* because of their high sample sizes and largely overlapping geographic distributions. If chloroplast mismatch was due to recent introgression, then the mismatched chloroplast should be closely related to geographically close samples from other species. For example, an *H. annuus* sample with a mismatched clade 1 chloroplast should be closely related to other Texas clade 1 chloroplast samples, as opposed to clade 1 chloroplast samples from California.

To test this, we identified each mismatched sample and found its most closely related nonmismatched sample in the phylogeny based on the topology (excluding branch lengths) and recorded the geographic distance between the target mismatched sample, and its closest nonmismatched relative. In some cases, multiple samples were equally related, in which case the average distance was calculated. To account for unevenness of species distributions and samples, we then compared this geographic distance with the distribution of geographic distances for all other possible nonmismatched samples. From this distribution, we calculate the percentile rank for the genetically closest relative. In this case, a value of 1 means that the genetically closest relative is also the geographically closest sample, and 100 means that it is the geographically furthest away. We expect that if ILS caused all mismatch, then the percentile scores should be a uniform distribution from 1 to 100 and have an average of 50, while introgression should produce values < 50. For each species, we then tested whether the percentiles were significantly different from 50 using a one-sample *t* test. This test was repeated using a version of the phylogeny with nodes collapsed if they had $\leq 50\%$ ultrafast bootstrap support.

To further test whether nuclear mismatch was due to recent admixture, we used Patterson's *D* to rank the relative admixed ancestry in each sample using nuclear SNPs (6). Nuclear SNPs were generated using the same pipeline as that of Todesco et al. (53), but included all species in joint genotyping. Briefly, sequence data were trimmed by Trimmomatic (v.0.36) (85), aligned to the *H. annuus* HA412v2 genome using NextGenMap (v.0.5.3) (86), sorted and duplicates removed using samtools (v.0.1.19) (87) and picardtools (v.2.9.3). Variants were called using GATK HaplotypeCaller (v.4.0.1.2) in windows along the genome and then filtered using GATK's VariantRecalibrator to retain SNPs above tranche = 90.0 (91). We additionally filtered the SNP table to retain variants with minor allele frequency > 0.01 and missing data less than 10%. Patterson's *D*, or the ABBA-BABA test, compares derived allele counts between two populations to ask whether one shares more derived alleles with a donor species, suggesting introgression. For this test, we used the perennial sunflower species as the outgroup. Prior analysis found that there were no early generation hybrids in the dataset (53) and because segregation is expected to rapidly unlink introgressed nuclear alleles and introgressed cytotypes in the absence of selection, any introgression-driven cytotypic mismatch is expected to be observed at the population level, rather than at the level of individual genomes. Therefore, we calculated Patterson's *D* using populations as groupings with Dsuite (92). We hypothesized that populations with a higher proportion of mismatched cytotypic will have more evidence of nuclear introgression than those that do not. We placed the target population in the P2 position, and then tested every other population of the same species in the P1 position. We then averaged the *D* score for all tests with the target population. In this way, the average *D* score is not measuring the absolute amount of introgression, but the relative amount of introgression compared to all other populations. For our five targeted taxa, *H. annuus*, *H. argophyllus*, *H. petiolaris petiolaris*, *H. petiolaris fallax*, and *H. niveus canescens*, we tested for introgression with all populations from possible donor species based on their major cytotypic. We classified populations as mismatched if they had more than one individual with mismatched cytotypic and used a two-sample *t*-test to compare the average *D* values between mismatched and nonmismatched populations.

Testing for Chloroplast Environment and Trait Associations. Previous work suggested that the chloroplast clade polymorphism was involved in local adaptation (52). To test for an association between environment and chloroplast or mitochondrial genome, we calculated the clade frequency for each population in *H. annuus*, *H. argophyllus*, *H. niveus canescens*, *H. petiolaris petiolaris*, and *H.*

petiolaris fallax, excluding all "other lineages" samples. For each population, we extracted environmental data (53, 93) for 39 variables, including 24 topoclimatic, 15 soil variables. Topoclimatic factors were extracted from climate data collected from 1961 to 1990 for the geographic coordinates of the population collection, while soil values were from soil collected during initial seed collections and published by Todesco et al. (53). Each value was Z-normalized. First, we tested for the effect of each variable independently with or without controlling for species using a general linear model with a quasibinomial variance model, in a type III ANOVA (94) (Eqs. 1 and 2). Within each type of test, we corrected each variables *P*-value for multiple testing using a Bonferroni correction.

$$\text{Proportion clade 1} \sim \text{species} + \text{environmental variable} \quad [1]$$

$$\text{Proportion clade 1} \sim \text{environmental variable} \quad [2]$$

To explore associations in multivariate space, we then used a PCA to identify the top five axes of environmental variation (PCs), representing 78% of total variation. For each of these PCs, we repeated the general linear model as before with PC score in the place of the environmental variable.

Finally, we divided the populations into species-specific sets and again used a general linear model to test for the effect of each environmental variable on chloroplast clade proportion to identify whether the direction of effect was consistent across species.

We then explored the effect of chloroplast clade on 91 phenotypic traits, as measured in a common garden (53, 55). We used trait data for 614, 299, 168, and 221 samples in *H. annuus*, *H. argophyllus*, *H. petiolaris petiolaris*, and *H. petiolaris fallax*, respectively. We excluded *H. niveus canescens* because it only had clade 1 and other lineages. Each trait was classified into one of four groups: architecture, inflorescence, leaf, or seed. Since many of the traits were correlated (e.g. multiple shape measurements for seeds), we used a PCA to extract the major axes of variation for each trait category. We then used the first two principal components in a linear model (PC score $\sim$ taxon + clade) to identify the effect of chloroplast clade on phenotype. We calculated percent variance explained by comparing the R^2 value of the full model with a reduced model excluding the clade. After finding significant effects on seed PCs, we further explored a linear model on each seed trait individually to isolate the specific traits affected.

Although chloroplast and mitochondrial clade were nearly perfectly linked, we repeated both trait and environment association tests using mitochondrial clade to see whether the associations were stronger or weaker.

Testing for Cytoplasmic-Nuclear Associations. We next aimed to detect whether natural selection was causing associations between nuclear alleles and cytotypic. This was complicated due to our sampling design. The samples used in this study were collected as seeds from the wild and grown in a benign common garden. In a population polymorphic for cytotypic, we expect that selection on cytonuclear interactions would cause linkage between cytotypic and nuclear variants, but this linkage would be reset after random mating (unless cytonuclear interactions affected mate choice or pollination success). Since our samples were subject to little selection after germination, linkage between cytotypic and nuclear loci is unlikely to be present. Because of this, we instead focused on population-level associations between cytotypic and nuclear allele frequencies. We expect that populations with more mismatched cytotypes would have allele frequency differences at interacting nuclear loci. This is similar to environment-genotype associations, which use population-level correlations between allele frequencies and an environmental variable. With this framework, we calculated Kendall's tau and its *P*-value between the cytotypic frequency and biallelic SNP frequency for each dataset. Biallelic SNP frequencies were taken from the study of Todesco et al. (53), and filtered to remove sites with a minor allele frequency < 5%. To identify outlier regions associated with cytotypic, we used WZA with 10 kb windows (59). This program was originally designed for genotype-environment associations. It combines the *P*-values for multiple SNPs within a window and identifies windows with disproportionately low *P*-values.

Although both our single SNP correlations and windowed WZA scans found highly significant putative cytotypic-SNP interactions, we were not satisfied that the null models were adequate. To test this, we permuted cytotypic frequencies by shuffling values between populations and reran the analysis 100 times for each species. We then visualized the log *P*-value scores for the upper percentiles in the permutations and the empirical values.

Data, Materials, and Software Availability. SRA accession numbers for all samples used are listed in [Dataset S1](#). The sequence data can be found in <https://www.ncbi.nlm.nih.gov/sra/PRJNA532579> (95), whereas all code is presented in https://github.com/owensgl/helianthus_cytoplasm_2024 (96).

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